VF Asia Footwear Laboratory Testing Manual

Test Method for Footwear Adhesion	Test method #: VF- F01 REF #: TM-R-032 Page 1 of 19
Date Issued: May, 2019 Revision #: 3 Reason for Revision: Rectify the unit of measure from kg to kgf for strength of material.	Created by: Nila Babia Erazo

PURPOSE AND SCOPE: To measure the peel strength of the bond between the outsole, midsole and upper materials in VF Footwear.

REFERENCE: ASTM D903

EQUIPMENT AND MATERIAL:

1. Tensile testing machine with data recording devise, preferably with chart recorder
2. Clamping jaws for the tensile tester
3. Razor or sharp knife.
4. Hot Knife and/or Hot Air Blower to initiate material separation
5. Clicker press or cutting table
6. Steel scale or calipers capable of measuring to 0.5mm
7. Footwear, or assembled midsole/outsole units.

SAMPLE PREPARATION:

1. Condition the sample for at least 48 hours at standard laboratory environment at 23.0±2.0 ˚C temperature and 55 ±10 % Relative Humidity to allow the cement to crosslink completely.
2. If the test sample is a whole shoe, the entire upper portion of the shoe must be removed by slicing with a razor blade or sharp knife along the midsole /upper joint. It will also be necessary to remove the heel counter, sock-liner and any threads used to stitch the upper to the outsole.
3. It is very important that care must be taken so that no damage to the bond being assessed.

I. Perimeter Pull Testing (Figures 3, 4, and 5)

1. For midsole to upper bond test, draw a line around the perimeter of the top side of the midsole/upper which has a width of 1.0+/-0.1cm from the edge. When the upper material bonds directly to the outsole in the top / toe area, the width must be reduced to 0.5+/-0.05cm.

2. For outsole to midsole bond test, draw a line around the perimeter of the bottom side of the midsole /outsole which has a width of 1.0+/-0.1cm from the edge.

3. Using a razor blade or sharp knife, accurately cut along this perimeter line so that the cut completely penetrates through the upper or outsole component and into the midsole.

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4. Once the perimeter cut has been made, it may be easier to initiate a peeling location by cutting through the upper or outsole material perpendicular to the perimeter. This cut should be made in the shank – medial area of the shoe, peel the 1.0cm wide upper or outsole strip away from the midsole for a distance of about 2cm to provide enough material for the clamps to grip.

   Note: In some cases, air blower and hot knife could be used to initiate separation between bonded materials by heating the beginning of the perimeter sample with a hot air blower to loosen the bond and separate with a knife of about 2 cm. Ensure proper ventilation to avoid exposure to fumes.

5. To have accurate spacing during testing, put a mark on the testing sample approximately every 2cm and number these locations starting at the initiation point in ascending order around the entire perimeter of the shoe.

6. Measure and record the nominal width of the perimeter test sample ***before testing***. This will be used to calculate force per unit width. ***(If the bottom perimeter for bonding area is NOT flat, may check the width after testing - the width to be checked is on the bonded area.)***

II. Area Pull Testing (Figures 6 through 9):

1. Using a band saw or sharp knife or cutting die, cut the section from the shoe or assembled test plaque. The specimens should be 1.0+/-0.1cm wide and at least 5-7.5cm long. If either the sole or midsole has a center cutout area, cut the specimen to the width of the cement margin of the sole. The width of the specimens should be cut as evenly as possible.

Note: ***Depending on the intent of testing, specimens may be pulled toe to heel, heel to toe, or from the midline of the shoe to the feather edge. "Toe Bumper" samples width will be dictated by the design. Some 'Toe Bumper' or direct attach soling constructions will be better suited to pulling apart the bonded material toward the upper/soling feather edge (see Figure 8). In this case the sample should be 2.5cm wide.***

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2. Initiate separation by cutting through the upper or outsole material perpendicular to the specimen, strip away for a distance of about 2cm to provide enough material for the clamps to grip.

Note: In some cases, air blower and hot knife could be used to initiate separation between bonded materials by heating the beginning of the perimeter sample with a hot air blower to loosen the bond and separate with a knife of about 2 cm. Ensure proper ventilation to avoid exposure to fumes.

3. To have accurate spacing during testing, put a mark on the testing sample approximately every 2cm and number these locations starting at the initiation point to the end of the specimen.

4. Measure and record the nominal width of the perimeter test sample ***before testing***. This will be used to calculate force per unit width.

III. Whole piece of unit sole bonded with 2 or more layers. (Figures 10-11)

1. If the sample is taken from a whole shoe, remove the entire upper carefully, by slicing with a razor blade or sharp knife along the midsole /upper joint. Ensure that the edge of the outsole attached to upper is not removed.

2. Initiate separation by cutting through the upper or outsole material perpendicular to the specimen, strip away for a distance of about 2cm to provide enough material for the clamps to grip.

3. To have accurate spacing during testing, put a mark on the testing sample approximately every 2cm and number these locations starting at the initiation point to the end of the specimen. Refer to figure 11

4. Measure and record the width of the test sample ***before testing***. This will be used to calculate force per unit width. Refer to figure 11.

REPETITIONS: Test at least one (1) upper to midsole bonds and, one (1) midsole to outsole bonds per shoe type.

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TEST PROCEDURE:

Test should be conducted at standard laboratory conditions @ 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

1. Equip the tensile testing machine with suitable jaws which are capable of clamping the test samples firmly in place such that no slippage occurs during testing.
2. Zero and calibrate the tensile testing machine. Set the crosshead speed to 100mm/minute and the load range to 20kgf, and a jaw gap of approximately 4cm.
3. Centrally and vertically clamp the test sample in the jaws of the tensile tester as follows:

Stationary Jaw		Moving Jaw
Rubber	to	Leather, Synthetic, Fabric
Rubber	to	EVA
Rubber	to	Leather, Synthetic, Fabric

Note: For other material combinations, the “stretcher” material is always clamped in the stationary jaw.

4. Operate the tensile testing machine at a crosshead speed of 100mm/minute.
5. As the test runs, monitor the bonded surfaces for material failure. If either surface is tearing rather than separating, use a sharp knife to cut back the tearing material to the bonding line until the two surfaces separate.
   - Do not use data which is produced while cutting is being performed.
   - Do not cut the test sample while bond delamination is occurring.
6. Do not allow the jaws to separate more than 30cm, before returning the jaws to their starting point and re-clamping the test sample.

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7. Record the load at each marked 2cm location. If chart recorder is available, plot the load value on the chart recorder at every 2cm marked.
8. As the test runs, the test sample should be marked in areas where cutting back to the line had occurred.
9. As the test runs, points falling below the standard should be marked on the test sample, and the type of bond failure should be recorded such as:

   AF - Adhesive failure or failure at the adhesive/substrate interface.

   CF - Cohesive failure or internal failure of the adhesive.

   SF - Substrate failure or tearing/breaking of one of the substrate materials.

   PF- Failure at the primer / substrate interface

10. Identify and record the maximum and minimum peak values on the chart recorder plot, to the nearest 0.1kgf (**L**).
11. Measure the width of the sample at the point which corresponds to each peak to the nearest 0.05cm (**W**).
12. Calculate the force required to pull the test specimen apart **at every 2cm marked** by dividing the force by the measured width of the test specimen. If an automated test system is used, this can be done automatically. For manual recording systems, the bonding strength should be calculated from individual points at 2 cm marked.

CALCULATION:

Peel Strength (kgf /cm) = L / W

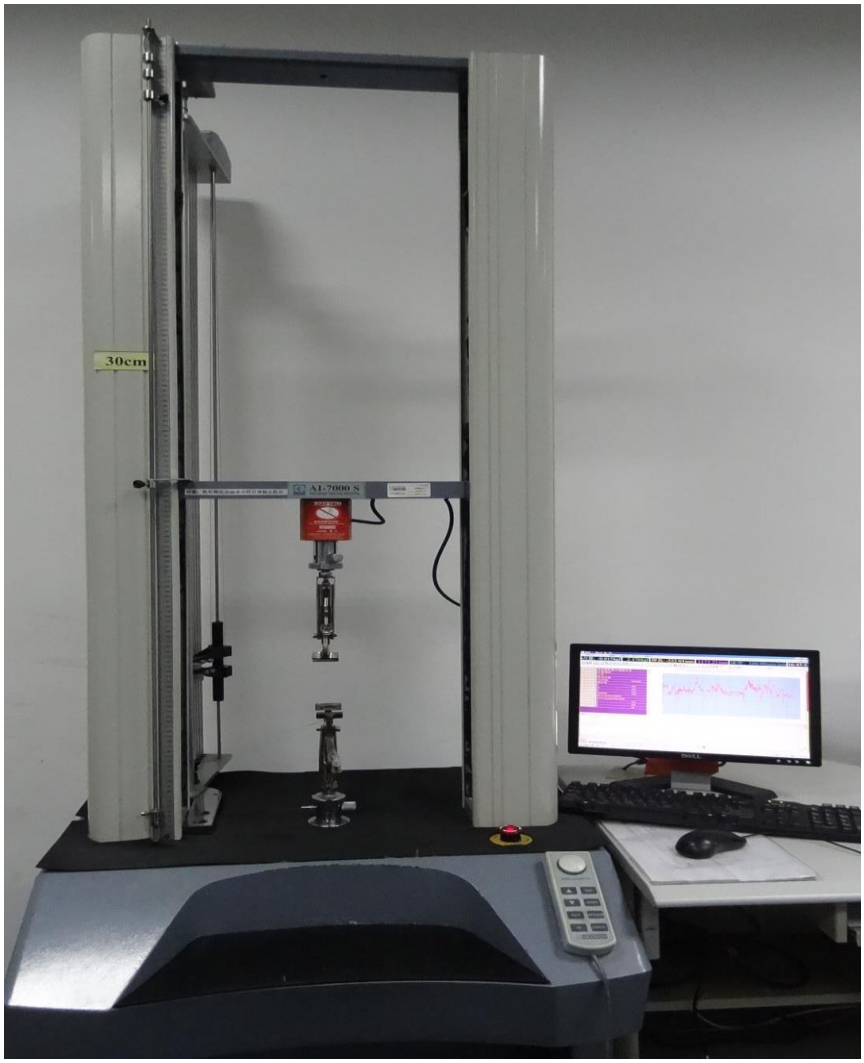
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TEST REPORT:

1. Bond strength - the **minimum** and **maximum** peaks to the nearest 0.1kgf /cm.
2. The mode of failure should be noted as follows:
    - **AF** = Failure at the adhesive /substrate interface of the adhesive/primer interface.
    - **CF** = Cohesive failure of the adhesive (failure within the adhesive or between the two layers of adhesive).
    - **PF** = Failure at the primer/substrate interface.
    - **SF** = Failure of the substrate, including the location of the failure within the substrate.
3. The reason for bond failure should also be reported , such as **Poor Priming , Poor cementing , Over dried cement , Wet Cement , Wet primer, Poor pressing , Poor roughing , Weak material , Dusty component**
4. Note any points in the toe, flex zone, or heel that drop below the standard for the shoe being tested.
5. Note any points not in the toe, flex zone, or heel, that drop below 75% of the standard for the shoe being tested.

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Sample of Test Equipment with Chart Recorder: Figure 1



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Clamping Jaws: Stationary Jaw (bottom) & Moving Jaw (up): Figure 2



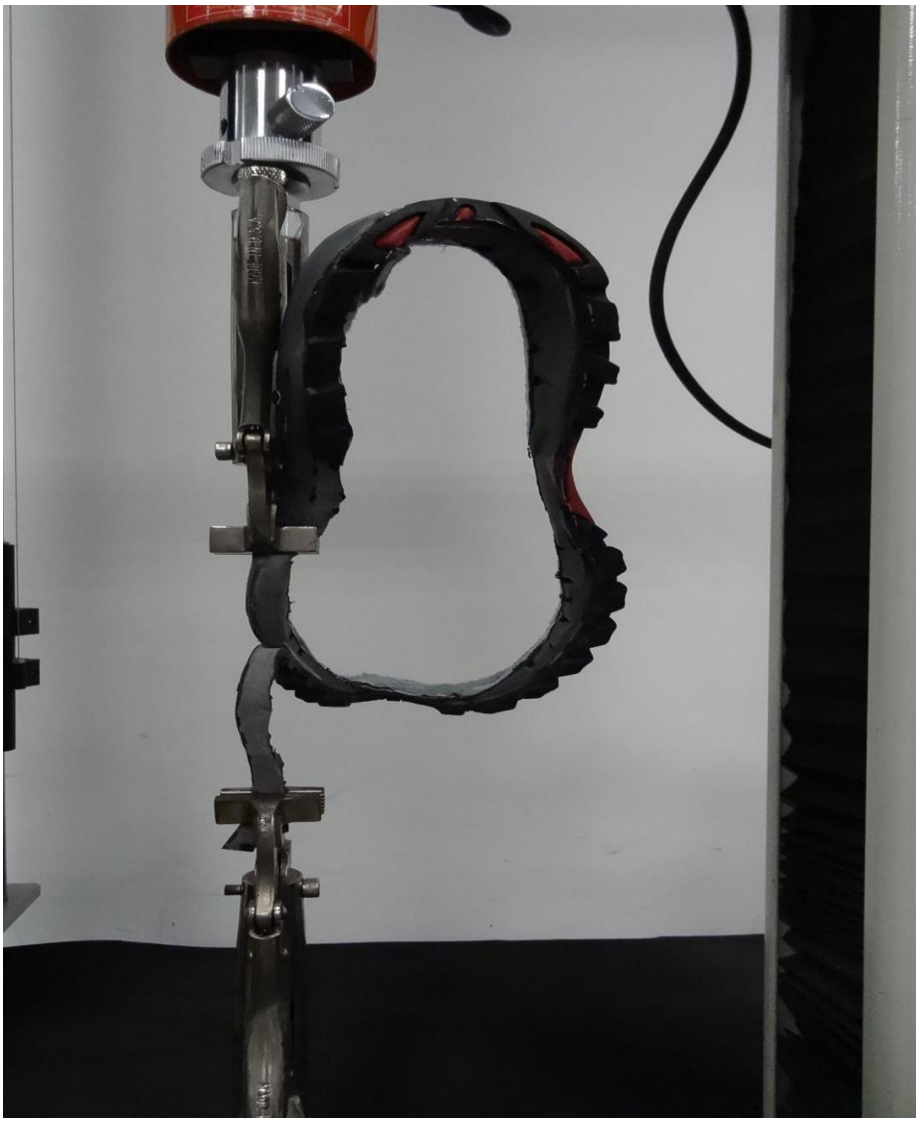
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Perimeter Pull Sample: Figure 3



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Outsole to Midsole Perimeter Pull Testing: Figure 4



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Midsole to Upper Perimeter Pull Testing: Figure 5



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Midsole to Outsole Area Pull Sample: Figure 6



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Midsole to Outsole Area Pull Testing: Figure 7



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Area Pull (Toe Bumper) Sample: Figure 8



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Outsole (Toe Bumper) to upper Area (Feather Edge) Pull Testing: Figure 9



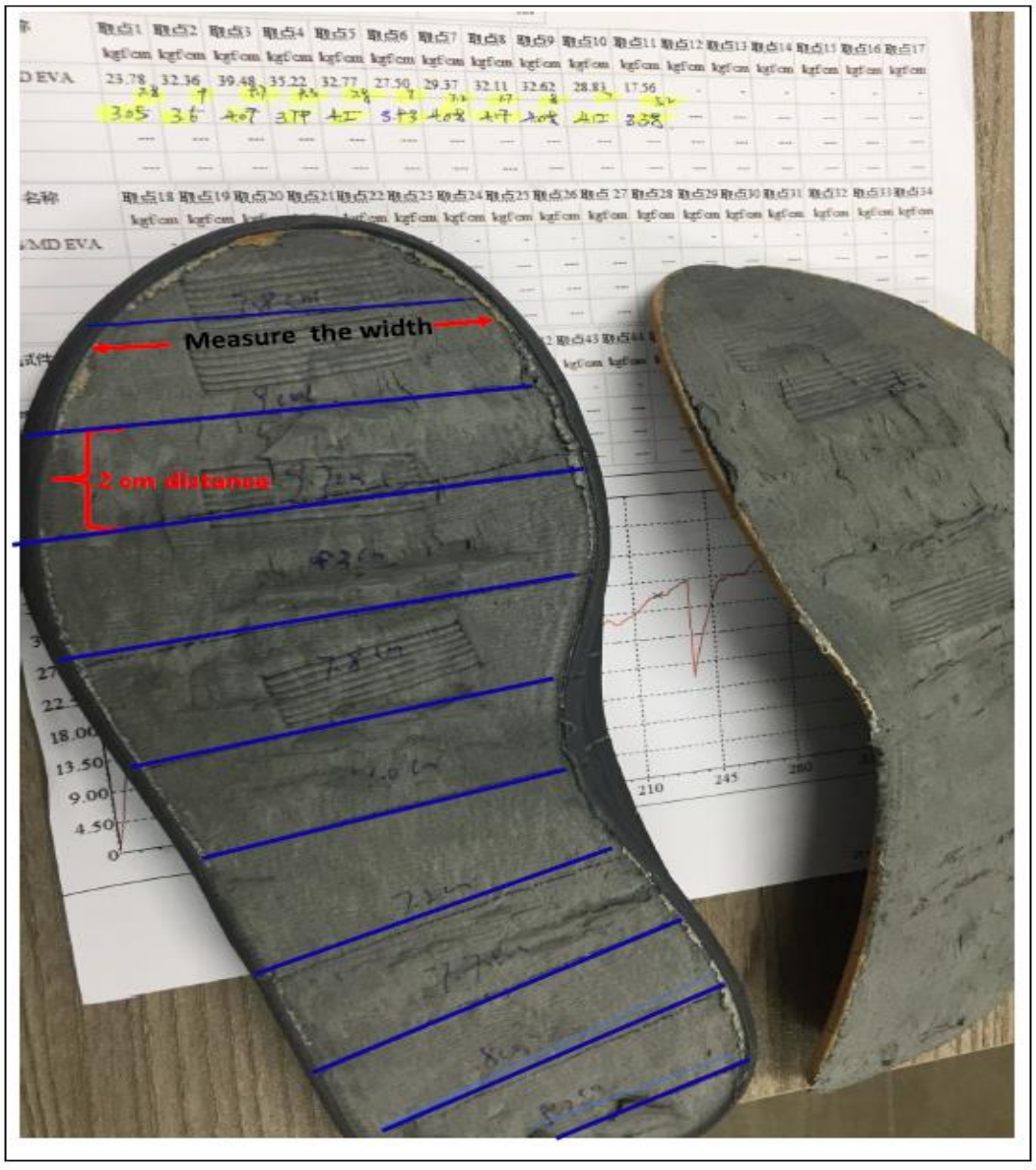
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Whole Piece of Outsole Bonded with other Component: Figure 10



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Test specimen marked 2 cm distance & Instruction to Measure the Width: Figure 11



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Sample of Graph for Bonding: Figure12

VF Bonding Test Result

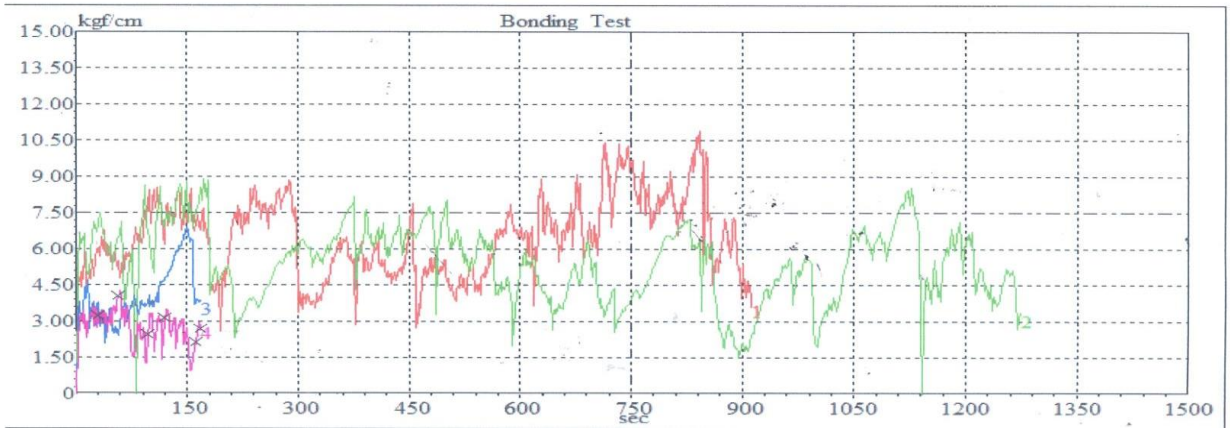
VFM 001-1

测试日期	测试速度 mm/min	试件编号	试件宽度 mm
2014-09-05	100.000	TNF-03-09-2014-BD01L	10.000
2014-09-05	100.000	TNF-03-09-2014-BD01R	10.000
2014-09-05	100.000	TNF-03-09-2014-BD01L	25.400
2014-09-05	100.000	TNF-03-09-2014-BD01R	25.400

试件名称	取点1 kgf/cm	取点2 kgf/cm	取点3 kgf/cm	取点4 kgf/cm	取点5 kgf/cm	取点6 kgf/cm	取点7 kgf/cm	取点8 kgf/cm	取点9 kgf/cm	取点10 kgf/cm	取点11 kgf/cm	取点12 kgf/cm	取点13 kgf/cm	取点14 kgf/cm	取点15 kgf/cm	取点16 kgf/cm	取点17 kgf/cm
Upper/toe port/MDEVA	5.342	5.879	6.521	8.049	7.610	6.916	4.555	7.245	7.774	7.720	3.756	5.528	5.594	5.834	4.901	5.274	5.349
OS/TPR/toe por/MDEV	6.683	7.083	6.077	7.728	4.471	4.474	3.968	6.140	5.949	8.064	6.324	6.710	6.363	6.475	6.379	4.444	4.436
OS/Insole	3.246	3.217	3.692	3.832	6.797	3.894	-	-	-	-	-	-	-	-	-	-	-
OS/Insole	3.254	4.051	2.469	3.126	2.132	2.693	-	-	-	-	-	-	-	-	-	-	-

试件名称	取点18 kgf/cm	取点19 kgf/cm	取点20 kgf/cm	取点21 kgf/cm	取点22 kgf/cm	取点23 kgf/cm	取点24 kgf/cm	取点25 kgf/cm	取点26 kgf/cm	取点 27 kgf/cm	取点28 kgf/cm	取点29 kgf/cm	取点30 kgf/cm	取点31 kgf/cm	取点32 kgf/cm	取点33 kgf/cm	取点34 kgf/cm
Upper/toe port/MDEVA	4.413	5.535	7.817	6.827	6.708	7.068	9.573	9.748	8.929	8.165	7.171	10.036	5.534	6.311	3.618	4.261	-
OS/TPR/toe por/MDEV	3.571	5.230	3.831	5.024	6.369	3.485	5.030	4.860	5.410	3.856	6.810	6.343	8.288	4.571	6.648	6.931	4.812
OS/Insole	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
OS/Insole	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

试件名称	取点35 kgf/cm	取点36 kgf/cm	取点37 kgf/cm	取点38 kgf/cm	取点39 kgf/cm	取点40 kgf/cm	取点41 kgf/cm	取点42 kgf/cm	取点43 kgf/cm	取点44 kgf/cm	取点45 kgf/cm	取点46 kgf/cm	取点47 kgf/cm	取点48 kgf/cm	取点49 kgf/cm	取点50 kgf/cm	取点51 kgf/cm
Upper/toe port/MDEVA	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
OS/TPR/toe por/MDEV	3.872	4.910	3.304	-	-	-	-	-	-	-	-	-	-	-	-	-	-
OS/Insole	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
OS/Insole	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-



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Sample of Test Report: Figure 13

Bonding Test Results
胶着力测试

SATRA Accredited Laboratory

Form # : Lab VF 001 | **Lab Ref.No: TNF 21-07-2014-BD02**

Factory (工厂):
Adhesive supplier胶水供应商:
Shipped Date (出货日期):
Model/Style/PGSC (英文鞋型): M Gannett Peak Clog Suede
Color (颜色): Demitasse Brown / Cub Brown
Production Date (生产日期):
PO/Lot No (订单号码): 600008100400001
PO Quantity (订单数量):
Send by (送达) : Simon / Factory CFA

O Pro. Shoes 量产鞋 O Pro. Re-test 量产重测
O New/Development 开发
o Tech 1 o Tech 2 o Tech 3 o Tech 4
O Sales Sample 销样
O Trial experiment 试胶试验
Received Date 收到日期: 7/21/2014
Tested Date 测试日期: 7/23/2014
Finished Date 完成日期: 7/23/2014
Technician 技术员: Polly Zhang

Shoe Part	VF Test Method	STD kgf/cm	Test Result kgf/cm Min	Test Result kgf/cm Max	Status Pass	Status Fail
A.Upper/ toe portion Left 鞋面/鞋头橡胶	VF-F01 (2019/05)	3.00	6.001	9.073	√	
A.Upper/ toe portion Right 鞋面/鞋头橡胶	VF-F01 (2019/05)					
C:Upper/MD EVA Left 鞋面/中底EVA	VF-F01 (2019/05)	3.00	3.111	5.731	√	
D:Upper/OS Right 鞋面/大底	VF-F01 (2019/05)					
E: OS /toe portion Left OS /鞋头	VF-F01 (2019/05)					
F:OS /toe portion Right OS /鞋头	VF-F01 (2019/05)	3.00	4.218	5.641	√	
G.OS /MD EVA Left 大底 / 中底 EVA	VF-F01 (2019/05)					
H. OS/MD EVA Right 大底/中底 EVA	VF-F01 (2019/05)	3.00	3.029	4.578	√	
I: Center: Insole/OS Left 中插 / 大底	VF-F01 (2019/05)	3.00	3.423	4.408	√	
J: Center: Insole/OS Right 中插 / 大底	VF-F01 (2019/05)	3.00	2.972	4.435	√	
K:Upper / TPR Left 鞋面 / TPR	VF-F01 (2019/05)					
L:Upper / TPR Right 鞋面 / TPR	VF-F01 (2019/05)					

Mode of Failure(s) 失败的方式

AF= Failure at the adhesive/substrate interface or the adhesive/primer interface

CF= Cohesive Failure of the adhesive (failure within the adhesive or between the two layers of adhesive).

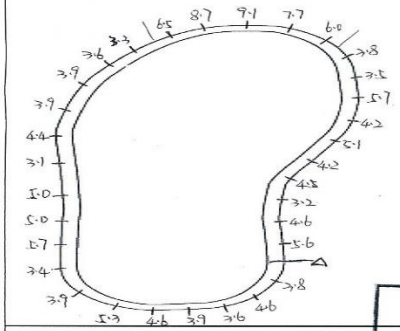
PF= Failure at the primer/substrate interface

SF= Failure of the substrate, including the location of the failure within the substtrate

Possible Cause(s) 可能的原因

(a) Poor Priming 药水不良
(b) Poor cementing 胶水不良
(c) Over-dried cement 积胶
(d) Wet primer 湿的药水
(e) Wet cement 湿的胶水
(f) Poor pressing 压机不良
(g) Poor Roughing 打粗不良
(h) Upper material - weak 鞋面材料不良
(i) EVA-Weak EVA 不良
(j) Dusty component 粉沫
(k) Splitting of foxing tape 围条分层

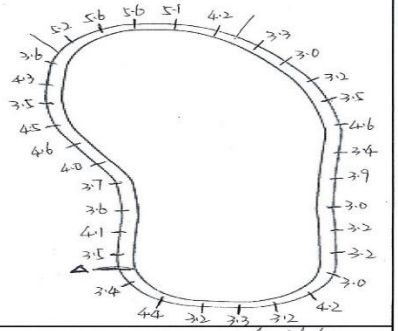
1.Upper / Toe portion / MD EVA (Left) kgf/cm



Center: IS/OS kgf/cm

Left	Right
3.6	3.0
4.2	3.4
3.4	3.6
4.4	3.9
4.4	4.4
3.7	3.8

2.OS / Toe portion / MD EVA (Right) kgf/cm



LABORATORY PASS

Comments 备注:

Reviewed by 审查:
Approved by 核准: